

SARAJEVO, Bosnia and Herzegovina

WMO: 133530

Lat: 43.825N

Lon: 18.331E

Elev: 521

StdP: 95.23

Time Zone: 1.00 (EUC)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.9	-9.9	-14.9	1.1	-11.4	-12.0	1.4	-8.8	9.5	10.1	8.0	10.2	0.9	290	0.475

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.7	33.1	19.7	31.2	19.6	29.2	19.2	21.6	29.4	20.6	28.3	19.9	27.2	2.7	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.1	14.7	25.3	18.1	13.9	23.8	17.2	13.1	22.6	65.3	29.2	61.9	28.2	59.0	28.0	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.0	6.2	4.8	-17.4	36.2	4.6	2.0	-20.7	37.7	-23.4	38.9	-26.0	40.0	-29.4	41.5		
			-16.8	23.4	4.4	1.0	-20.0	24.1	-22.6	24.7	-25.0	25.3	-28.3	26.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.6	-0.3	1.6	5.9	10.6	14.8	18.9	20.7	20.9	15.8	11.3	6.2	0.5
	DBStd	8.47	4.91	4.94	4.37	3.62	3.31	3.54	3.09	3.01	3.52	3.99	4.77	4.82
	HDD10.0	1197	320	238	137	34	4	1	0	0	2	34	132	295
	HDD18.3	3067	578	470	386	232	114	36	14	12	89	220	365	552
	CDD10.0	1420	1	1	9	53	153	266	331	337	177	73	17	2
	CDD18.3	248	0	0	0	0	6	52	86	90	14	1	0	0
	CDH23.3	3003	0	0	1	23	152	625	991	999	201	11	0	0
	CDH26.7	1167	0	0	0	2	24	216	424	441	59	1	0	0
Wind	WSAvg	1.5	1.3	1.6	1.9	1.8	1.5	1.5	1.5	1.4	1.3	1.1	1.5	1.4
Precipitation	PrecAvg	1049	74	76	75	84	84	94	82	71	101	104	109	97
	PrecMax	1508	179	184	160	179	187	216	148	177	263	215	217	268
	PrecMin	760	15	14	4	16	44	12	20	8	22	8	19	0
	PrecStd	197	40	43	41	38	36	45	35	43	62	51	45	58
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.9	17.0	21.0	26.1	29.1	33.0	35.9	36.0	31.2	24.9	20.0	15.0
		MCWB	9.4	10.1	10.9	13.9	17.6	20.3	19.5	19.5	17.9	16.5	13.4	11.2
	2%	DB	11.8	13.9	18.2	23.1	27.0	31.1	33.1	33.2	28.0	22.8	18.0	12.9
		MCWB	8.4	8.5	10.0	13.3	16.9	19.9	19.8	19.4	17.7	15.7	12.9	9.2
	5%	DB	9.0	11.8	16.1	20.8	25.0	29.2	31.1	31.1	25.2	20.2	16.1	10.2
		MCWB	6.6	7.8	9.3	12.1	16.0	19.5	19.8	19.5	17.1	14.6	11.6	7.6
	10%	DB	6.2	9.2	14.0	18.2	22.9	27.1	28.9	29.0	23.0	18.1	14.1	8.0
		MCWB	5.0	6.3	8.6	11.2	15.4	18.7	19.4	19.2	16.2	13.8	10.5	6.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.7	10.6	12.1	15.2	19.1	22.6	22.7	22.8	19.8	17.6	14.5	11.5
		MCDB	12.9	15.1	18.5	23.5	26.4	29.8	30.5	30.5	26.3	23.1	18.3	14.4
	2%	WB	8.6	9.1	10.8	13.8	17.8	21.2	21.5	21.4	18.6	16.3	13.1	9.4
		MCDB	11.4	13.5	16.6	21.0	24.6	28.9	29.5	29.1	25.6	20.9	17.2	12.1
	5%	WB	6.8	7.9	9.8	12.8	16.7	20.1	20.6	20.4	17.7	15.2	12.0	8.2
		MCDB	8.8	11.5	15.1	18.8	23.1	27.3	28.4	28.0	24.1	19.2	15.7	10.3
	10%	WB	5.1	6.4	8.9	11.9	15.7	19.1	19.8	19.6	16.8	14.2	10.7	6.4
		MCDB	6.4	9.0	13.8	17.1	21.6	26.0	27.2	26.9	21.9	17.6	13.9	7.9
Mean Daily Temperature Range	5% DB	MDBR	7.4	9.2	11.0	12.2	12.9	13.0	13.7	14.4	12.2	11.4	8.8	6.7
		MCDBR	10.9	13.5	16.3	17.7	17.4	16.8	18.1	18.9	16.9	15.4	12.2	9.7
		MCWBR	7.7	8.7	9.0	8.7	8.4	7.5	7.1	7.2	7.7	8.8	7.6	7.0
	5% WB	MCDBR	10.0	12.3	14.1	14.5	15.5	15.6	15.6	15.9	14.2	13.2	11.1	9.0
		MCWBR	7.3	8.1	8.1	7.7	7.9	7.4	7.0	7.0	6.9	7.8	7.2	6.6
Clear-Sky Solar Irradiance	taub	0.320	0.351	0.398	0.454	0.449	0.441	0.441	0.442	0.416	0.381	0.343	0.319	
	taud	2.373	2.252	2.175	2.059	2.123	2.203	2.203	2.197	2.245	2.334	2.390	2.390	
	Ebn at Noon	798	827	830	810	827	834	829	813	803	785	762	761	
	Edh at Noon	79	105	130	158	152	141	140	136	120	97	78	71	
All-Sky Solar Radiation	RadAvg	1.44	2.14	3.31	4.25	5.26	6.03	6.25	5.63	3.93	2.73	1.66	1.23	
	RadStd	0.20	0.26	0.45	0.55	0.47	0.51	0.48	0.53	0.39	0.31	0.20	0.22	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (29 neighbors)	+0.51	N/A	N/A	N/A	N/A	N/A	N/A	-106	+112	N/A

Nomenclature: See separate page